

Assignment: #2

Unit: 5

Name: Kavshal kumar Raj

PRN: 240205241023

Branch: B.Tech CSE with AI & ML

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Subject: Deep Learning

① What is image processing with suitable examples in context of DL.

→ Image processing in DL refers to using neural networks, specially convolutional Neural Networks (CNNs) to automatically analyze, enhance, and interpret images. Instead of manually designing features, the model learns patterns such as edges, shapes, textures, & object identity from raw pixel data.

② Mention two applications of deep learning in image processing.

- Face recognition: used in smartphones, & security systems to identify individuals.
- Medical imaging: detecting tumors or diseases from X-rays & MRI's or CT scans.

③ What is Natural Language Processing (NLP)?

→ Natural Language Processing (NLP) is a field of artificial intelligence that enables computers to understand, interpret and generate human language. It allows machines to work with text and speech data for tasks like translation, chatbots, and sentiment analysis.

④ What are the major challenges in NLP?

→ NLP faces challenges such as ambiguity in language, understanding context, handling multiple languages and dialects, sarcasm detection, and dealing with incomplete or informal text.

⑤ What is Speech Recognition?

→ Speech recognition is a technology that converts spoken language into written text. It works by analyzing audio signals, extracting features, and using trained models to recognize words and sentences.

⑥ What is the difference between speech recognition & speaker recognition?

→ Speech recognition identifies the content of speech (what is being said), while speaker recognition identifies the identity of the person speaking based on voice characteristics.

⑦ What is video analytics?

→ Video analytics is the use of artificial intelligence to automatically analyze video streams and detect events, objects or behaviours. It reduces the need for continuous human monitoring.

⑧ How is video data different from image data?

→ Image data is static and contains information from a single frame. Video data consists of a sequence of frames over time, including temporal information such as motion and object tracking.

⑨ What role do CNNs play in image processing applications?

→ CNNs extract spatial features from image using convolution and pooling operations. They are highly effective for tasks like image classification, object detection, and image segmentation.

⑩ What is tokenization in NLP?

→ Tokenization is the process of breaking text into smaller units called tokens, such as words, subwords, or sentences. These tokens are then converted into numerical form for model processing.

⑪ What is sentiment analysis?

→ Sentiment analysis is an NLP task that determines the emotional tone of a text, such as positive, negative, or neutral. It is commonly used in product reviews and social media analysis.

⑫ what is automatic speech recognition (ASR)?

→ Automatic Speech Recognition (ASR) is a system that automatically converts spoken audio into text using deep learning models trained on large speech datasets.

⑬ what is object detection in videos?

→ object detection in videos involves identifying and locating objects in each frame and often tracking their movement across multiple frames over time.

⑭ what is feature extraction in speech processing?

→ Feature extraction in speech processing converts raw audio signals into meaningful numerical representations, such as MFCCs, that capture important speech characteristics for recognition tasks.

⑮ List two real-world applications of video analytics.

- • Smart traffic monitoring systems
- Security surveillance for detecting suspicious activities.

Long-questions

① Explain the role of D.L in image processing with suitable example

→ DL plays a major role in image processing by automatically learning visual features from raw pixel data. Instead of manually designing features, models like CNNs extract edges, shapes, textures, and high-level object representations.

Key Roles:

- Image classification: Identifying object in images.
- Object detection: locating objects within images.
- Image segmentation: Dividing an image into meaningful regions.
- Face Recognition: Identifying individuals from images.

Examples:

- medical diagnosis using MRI scans.
- self-driving cars detecting pedestrians and traffic signs.

Deep learning improves accuracy because it learns hierarchical features layer by layer.

② Describe various deep learning techniques used in image classification and object detection.

→ For Image Classification:

- Convolution Neural Networks (CNNs) - Extract spatial features.
- Pretrained models like ResNet or VGG improve performance using transfer learning.

For Object Detection:

- R-CNN family: Region-based detection.
- YOLO (You only look once) - Real-time object detection.
- SSD (Single Shot Detector) - Fast and efficient detection.

These models classify objects and also predict their bounding box locations.

③ Explain Natural Language Processing (NLP) and discuss its major applications.

→ Natural Language Processing (NLP) is a branch of AI that enables computers to understand, interpret, and generate human language.

Major Applications:-

- Machine Translation (eg language conversion)
- Sentiment Analysis
- Chatbots and virtual Assistants
- Text Summarization
- Question Answering systems.

Deep learning models like RNNs and Transformers significantly improved NLP performance.

④ Describe the pipeline of an NLP system,
From text input to output.

→ NLP Pipeline:

1) Text input

2) Preprocessing

- Tokenization
- Stop-word removal
- Stemming / Lemmatization

3) Feature Representation

↳ word embeddings
(word2vec, etc)

4) Model processing

↳ Neural Networks
(RNNs / Transformers)

5) Output Generation

↳ classification or text generation

Here,

Each stage prepares raw text for meaningful machine understanding.

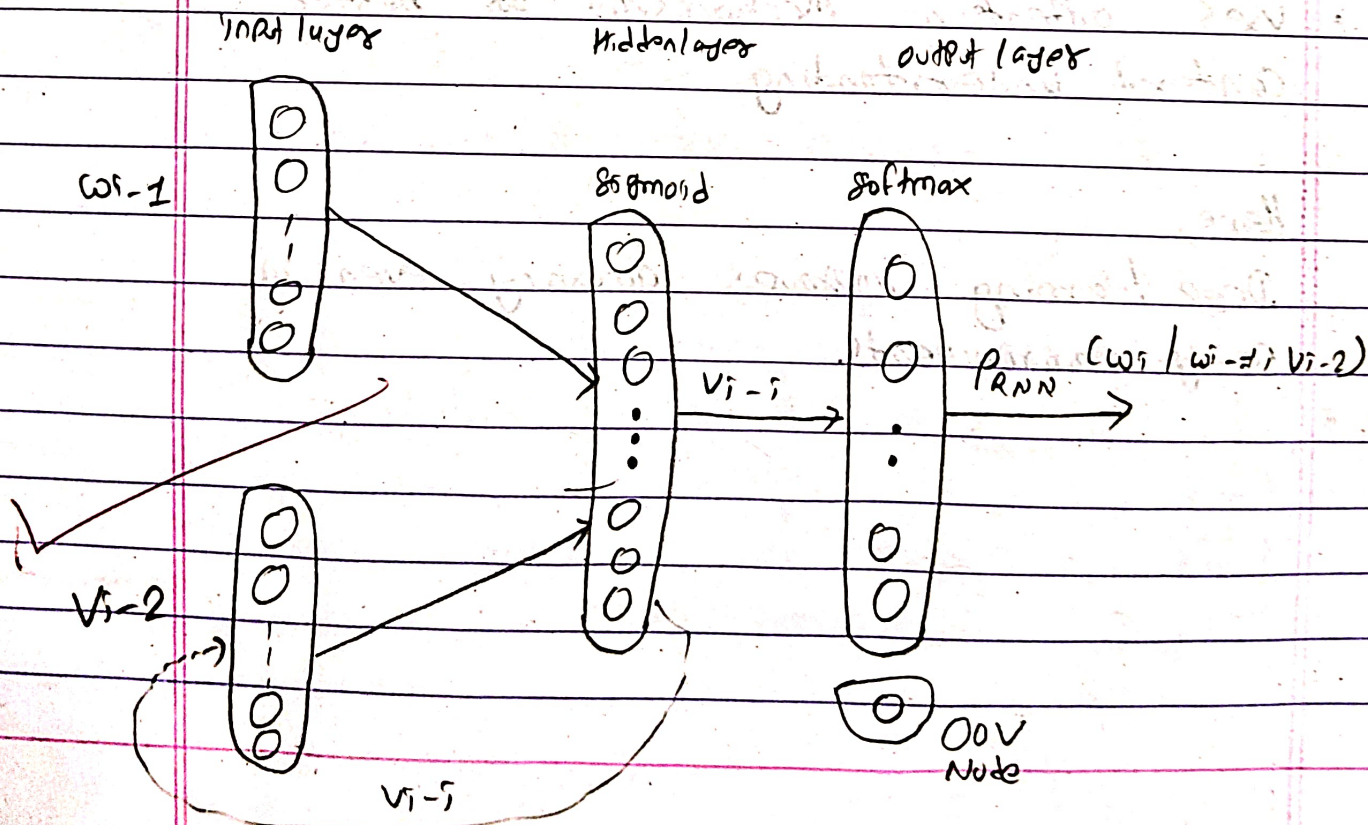
⑤ Explain how deep learning models are used in speech recognition systems.

→ Speech recognition systems convert spoken audio into text using deep learning.

Process:-

- i) Audio signal is converted into features (e.g., spectrogram or MFCC).
- ii) Deep models (RNNs, LSTM, CNNs) process time-based patterns.
- iii) output layer predicts words or characters.

Modern systems use deep neural networks for accurate and real-time transcription.



⑥ Discuss the challenges involved in speech recognition and how deep learning addresses them.

→ Challenges:

- Background noise
- Different accents
- Speaking speed variations
- Homophones

How deep learning helps?

- Learns robust features from large datasets
- Handles sequential data using RNN/LSTM
- Uses attention mechanisms for better context understanding.

Hence,

Deep learning improves accuracy even in noisy environments.

⑦ Explain video analytics and discuss its applications in real-world scenarios.

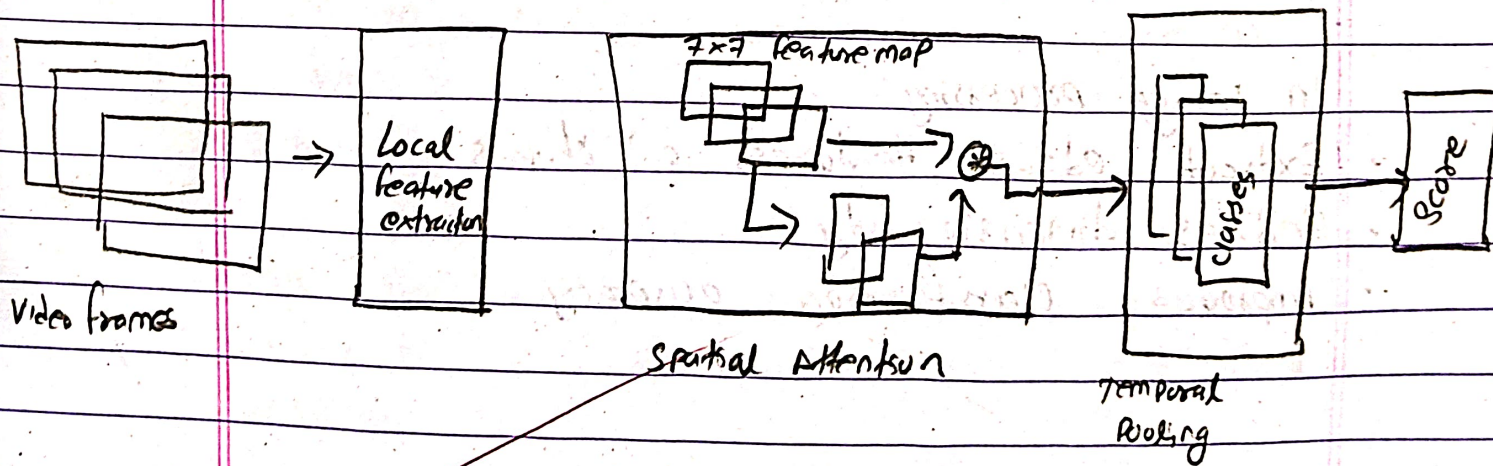
→ Video analytics uses AI models to automatically analyze video streams to detect events, objects or behaviours.

Applications:-

- Traffic monitoring system
- Smart Surveillance systems
- Retail customer behaviour analysis
- Sports performance tracking

It combines image processing with temporal analysis for intelligent decision-making.

⑧ Describe how CNNs and RNNs are used together for video analysis tasks.



In video analysis:

- CNNs extract spatial features from each video frame
- RNNs (or LSTMs) process these features across time to capture motion patterns.

This combination is useful for:-

- Action recognition
- Video captioning
- Activity detection

CNN handles "what is in the frame" while RNN handles "what is happening over time."

⑨ Explain the importance of feature extraction in image and speech processing.

→ Feature extraction converts raw data into meaningful representations.

In Image processing:

- Extract edges, textures, and shapes
- Reduces dimensionality
- Improves classification accuracy

In speech processing:

- Converts raw audio into MFCC or spectrogram features.
- Captures important frequency information.

Good feature extraction improves models performance and efficiency.

(10) Discuss ethical and privacy concerns related to deep learning applications in vision, speech, and language.

→ Deep learning applications raise important concerns:-

Privacy Issues:-

- Facial recognition without consent.
- Voice data recording.
- Surveillance misuse.

Ethical Issues:-

- Bias in datasets leading to unfair predictions.
- Misuse in deepfakes.
- Lack of transparency in decision-making.

Responsible AI development requires fairness, accountability, and proper data protection policies.